Introduction to Deep Learning

Neural Networks

A neural network is made of layers of simple units called neurons. Each neuron takes a weighted sum of its inputs, adds a bias and passes the result through an activation function such as ReLU. Stacking many layers lets the network learn complex patterns.

Training with Backpropagation

Networks are trained by comparing their predictions with the correct answers using a loss function. Backpropagation computes how much each weight contributed to the error, and an optimiser such as stochastic gradient descent or Adam updates the weights a little at a time.

Convolutional Networks

Convolutional neural networks, or CNNs, are designed for images. Small filters slide across the image to detect edges, textures and shapes, and deeper layers combine these into whole objects. CNNs are widely used for image classification and medical imaging.

Transformers

Transformers process sequences using a mechanism called attention, which lets every token look at every other token. They power modern large language models and have largely replaced recurrent networks for text tasks.